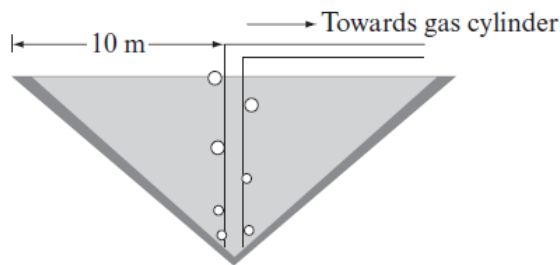


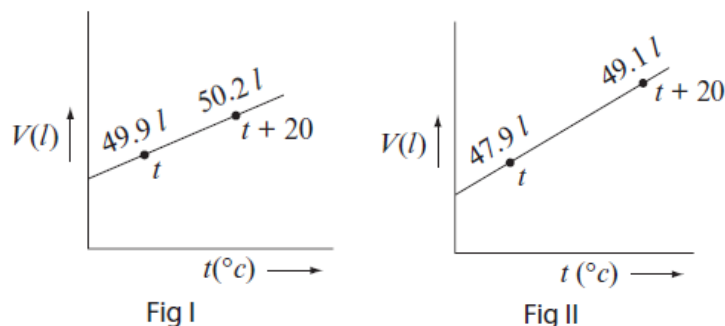
ASSIGNMENT (Subjective) (Gas Law)

1. A swimming pool (Figure) is conical in shape of diameter 20 m. To measure the depth of the pool, a pipe line having cross sectional area $\pi \text{ mm}^2$ has been introduced through the base centre to the depth of the pool. Other end of the pipe line is connected with the gas cylinder. When gases are passed through the pipe line, it has been measured that the volume of each bubble across the pipe line at the surface of the pool is $2\pi \text{ mm}^3$. Assume the temperature of water is same as that of atmosphere and 1 atm pressure is equal to 10 m of water column. If the water holding capacity of the swimming pool is $V \text{ m}^3$, then the value of $3V/100\pi$ is

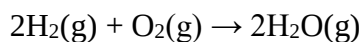


2. An iron meteorite was analysed for its isotopic argon content. The amount of Ar^{36} was 1.911 mm^3 (STP) per kg of meteorite. If each Ar^{36} atom had been formed by a single cosmic event, how many such events must there have been per kg of meteorite? If the answer is X, then the value of $X/10^{16}$ is ($R = 8.4 \text{ J/K-mol}$, $N_A = 6.0 \times 10^{23}$)
3. What will be the temperature difference (in $^{\circ}\text{C}$) needed in a hot air balloon to lift $10.0/8.314 \text{ kg}$ of mass? Assume that the volume of the balloon is 91.0 m^3 , the temperature of the ambient air is 17°C , the pressure is 1.0 bar, and air is an ideal gas with an average molar mass of 29 g/mole .
4. A vessel of volume 3.0 l contains ideal gas at the temperature 0°C . After a portion of the gas has been let out, the pressure in the vessel decreases by 0.8 atm (the temperature remaining constant). The gas density under the normal condition (0°C and 1 atm) is 1.25 g/l . The mass of released gas in (g) is
5. A 5.0 l box contains 28.4 g of mixture of gases C_xH_8 and C_xH_{10} . The pressure of the gas mixture at 300 K is 2.46 atm . The analysis of the gas mixture shows that carbon is 84.5% , by mass. The value of x is

6. An ideal gas, X_n , is taken in a container connected to frictionless, massless piston. The pressure of gas is constant (always equal to the atmospheric pressure). The expected graph for the variation of volume of gas with the change in temperature was to be like fig. I, but the actual graph is like fig II. The deviation was attributed to dissociation of gas as: $X_n(g) \rightarrow nX(g)$. If the yield of this reaction is only 60%, the value of n is



7. A mixture of H_2 and O_2 in 2:1 mole ratio is used to prepare water vapour by the reaction:



The total pressure of gases in the container is 4.5 atm at $57^{\circ}C$, before the reaction. The final total pressure of gases (in atm) at $127^{\circ}C$ after reaction assuming 80% yield of water vapour is

8. What would be the final pressure of O_2 in the following experiment? A collapsed polyethylene bag of 30 l capacity is partially blown up by the addition of 10 l of N_2 at 0.9 atm at 298 K. Subsequently, enough O_2 is pumped into the bag so that at 298 K and external pressure of 1.1 atm, the bag contains full 30L. If the answer is x atm, then the value of $10x$ is
9. Pure water vapour is trapped in a vessel of volume 10 ml. The relative humidity is 40%. The vapour is compressed slowly and isothermally. The volume of vapour (in ml) at which first drop of liquid will form, is
10. A cylinder contains 64 g of an ideal gas ($M = 64$) at $27^{\circ}C$ and 3 atm. In transportation, the cylinder fell down and a dent was created, i.e., the effective volume of the cylinder decreases. But the valve attached to the cylinder cannot keep the pressure greater than 3 atm, so 8 g of gas leaked out. The volume of the cylinder before and after the dent was a l and b l, respectively. If the valve were pretty strong, the pressure of gas in the cylinder after dent were $cd/7$ atm. Temperature remained constant during this process. The value of $abcd$ is

- 11.** A diver at a depth of 10 m exhales some air by which a bubble of air of volume 24.0 ml is formed. The bubble catches an organism which survives on the exhaled air trapped in the bubble. The organism just inhales the air at the rate of 0.05 millimoles/min and exhales nothing. The atmospheric pressure is 1 atm and the temperature of water is throughout 300 K. Density of water is 1.013 g/ml. ($R = 0.08 \text{ l-atm/K-mol}$, $g = 10 \text{ ms}^{-2}$)

ab = volume (in ml) of bubble when it reaches the surface after 10 min

cd = average rate (in 10^{-5} mole/min) at which the organism should inhale air so that volume of bubble remains the same at the depth and at the surface. The value of abcd is
- 12.** A 50 litre vessel is equally divided into three parts with the help of two stationary (fixed) semi-permeable membranes (SPM). The vessel contains 60 g H_2 in the left chamber, 160 g O_2 in the middle and 140 g N_2 in the right chamber. The left SPM allows transfer of only H_2 gas while the right SPM allows the transfer of both H_2 and N_2 . If the ratio of final total pressures in the three chambers is $x:y:5$, then the value of $(5 \times x \times y)$ is
- 13.** Under the identical conditions of temperature, the density of a gas X is two times to that of gas Y while molecular mass of gas Y is three times that of X. Calculate the ratio of pressure of X and Y.
- 14.** If excess $\text{F}_2(\text{g})$ reacts at 150°C and 1.0 atm pressure with $\text{Br}_2(\text{g})$ to give a compound BrF_n , if 423 mL of $\text{Br}_2(\text{g})$ at the same temperature and pressure produced 4.2 g of BrF_n , what is n ? [Atomic mass Br = 80, F = 19]
- 15.** Initially bulb 'A' contained oxygen gas at 27°C and 950 mm of Hg and bulb 'B' contained neon gas at 27°C and 900 mm of Hg. These bulbs are connected by a narrow tube of negligible volume equipped with a stopcock and gases were allowed to mix-up freely. Then obtained pressure in the combined system was found to be 910 mm of Hg.